

Lattice Before Ring: A Two-Level Silicon Photonic Demultiplexer, Benchmarked and Tested for Fabrication Tolerance

Zhongwei Wang

edX public username: wzwqoo · SiEPIC openEBL, 220 nm SOI, three cells of $605 \times 410 \mu\text{m}$

Abstract: This is the short version of the design report, written for peer review. Three layouts were submitted, each with a separate job: All three pass the EBeam DRC and SiEPIC verification with zero errors. Every device was simulated in Lumerical, and its expected spectrum is given here.

- **Chip 1 (EBeam_wzwqoo_design):** a two-level demultiplexer. An MZI lattice tree splits four coarse channels 10 nm apart, and microrings followed by contra-directional couplers (CDCs) pick four fine 200 GHz channels.
- **Chip 2 (EBeam_wzwqoo_benchmark):** the same coarse job done three other ways (an AWG, an echelle grating, a cascade of CDCs), plus a Fabry–Pérot cavity.
- **Chip 3 (EBeam_wzwqoo_FabChallenges):** copies of the chip-1 devices drawn deliberately too narrow or too wide, to measure how width errors move each filter.

Keywords: wavelength-division multiplexing, MZI lattice filter, microring resonator, contra-directional coupler, arrayed waveguide grating, fabrication tolerance

1. Summary for Reviewers

Table 1. Where each review criterion is answered

Criterion	Short answer	Section
Design concept	One system (chip 1), three alternatives to compare it with (chip 2), and width-offset copies to test its weak point (chip 3). 18 devices in total	2, 3, 4
Parameter variation	Four ring radii, two CDC periods, three cascade periods; width offsets $-15 / +8 / +15$ nm on rings, CDC-A and MZIs; two delay-guide widths (0.5 and 3 μm); two ring widths (500 and 479 nm)	3, 7
Calibration	GC loopback GCLoopback (chip 1); a stand-alone CDC-A (CDCA) to separate the CDC from the rings; the chip-1 ring and CDC-A are the 0 nm references for chip 3	6
Manufacturability	EBeam DRC: 0 errors on all three chips. PCell parameters were changed where the defaults would break the 70 nm Si spacing	5
Mask layout	One top cell per file, all shapes inside the floorplan, no waveguides closer than 2 μm except two harmless 1.7 μm runs on chip 2	5
Design for test	Every device: input on the opt_in grating coupler, outputs on the same column at 127 μm pitch, all labels unique	6
Submission	File name = top cell name: EBeam_wzwqoo_design, EBeam_wzwqoo_benchmark, EBeam_wzwqoo_FabChallenges	5

2. Design Concept

2.1. The problem

Optical links carry many wavelengths on one fibre, and the receiver must separate them. Two kinds of filter are needed:

- a **coarse** filter that splits wide bands (here 10 nm apart) with a **flat** top, so everything inside a band passes equally;
- a **fine** filter that picks single channels 200 GHz (1.6 nm) apart with a **narrow** line.

No single filter does both well. An AWG, for example, would need at least 50 channel slots and star couplers larger than the 605 μm cell to cover both grids.

2.2. The two-level architecture (chip 1)

- **Coarse level: a 1×4 MZI lattice tree (D1).** Each stage is a chain of unbalanced MZIs. Its coupler ratios shape a flat-topped passband 8.57 nm wide, wider than the 4.78 nm the fine channels occupy.
- **Fine level: four add-drop microrings (D2).** Each ring ($R \approx 7.5 \mu\text{m}$) drops one 200 GHz channel with a 0.2 nm wide line. Its FSR (12.1 nm) is larger than the 10 nm coarse spacing, so in the full system the lattice in front removes the ring's other orders.
- **Order selector on the test chip: CDCs.** On chip 1 the rings are measured without the lattice in front. A CDC after each ring passes a single band (it has no FSR), so the ring's other orders are removed on the chip as well.

Why a lattice first, and not an AWG? In simulation, an AWG coarse stage followed by fine filters loses the two outer fine channels by about 2 dB more than the inner ones. An AWG's passband is rounded (Gaussian) and only 3.0 nm wide at 1 dB, narrower than the 4.78 nm the fine channels span. The lattice's flat top covers all four.

2.3. Wavelength plan

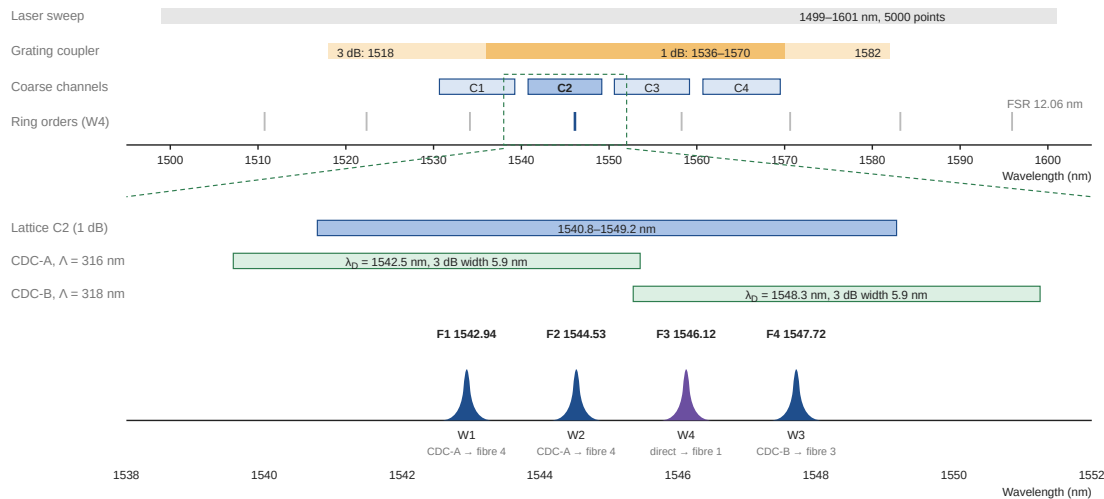


Fig. 1. Wavelength plan against the test window. Top: laser range, grating-coupler band, the coarse channels C1–C4, and the other orders of one ring. Bottom: zoom on C2 with the CDC bands and the fine channels F1–F4.

- **Coarse channels C1–C4: 1535, 1545, 1555, 1565 nm.** The spacing is 10 nm, not the 20 nm CWDM grid, because the grating couplers lose too much at the band edges: a 20 nm grid would lose 9.96 dB on its outer channels, against 2.56 dB here.
- **Fine channels F1–F4: 1542.94, 1544.53, 1546.12, 1547.72 nm.** They are on the ITU 200 GHz grid, inside C2, the flattest part of the grating-coupler response.
- **Fine linewidth: 0.2 nm.** It gives about 10 laser points per line with the 20.4 pm test-laser step, and –24 dB between neighbouring channels.

2.4. What each chip tests

Table 2. The 18 devices

Chip	Label	Device	Question it answers
1	LatticeD1	1×4 MZI lattice tree	Are the coarse bands flat and isolated?
1	RingCDCD2	4 rings + CDC-A + CDC-B	Do the rings land on F1–F4, and does the CDC remove their other orders?
1	CDCA	CDC-A alone	Where is the CDC band, without the rings?
1	GCLoopback	two GCs joined by a waveguide	Grating-coupler response, to normalise every spectrum
2	AWGC	1×4 arrayed waveguide grating	Benchmark against the lattice
2	EDGC	1×4 echelle grating, Bragg-mirror facets	Benchmark against the lattice
2	CDCCascade	three apodised CDCs on one bus	Benchmark against the lattice
2	FPCavity	two Bragg mirrors + half-period spacer	Can a single narrow line separate two neighbouring fine channels?
3	MZWireM15, MZWireP15	MZI, delay in 500 nm wire, width –15 / +15 nm	How far does a width error move the fringes?
3	MZWideM15, MZWideP15	the same MZI, delay in a 3 µm guide	Does a wide delay guide remove that drift?
3	CDCAM15, CDCAP8	CDC-A, width –15 / +8 nm	How far does the CDC band move?
3	RingM15, RingP8	the F2 ring (500 nm), width –15 / +8 nm	Does the ring line move with the CDC band, or away from it?
3	RingMatchM15, RingMatchP8	F2 ring redrawn at 479 nm width	Does a ring of matched width stay inside the CDC band?

The label column is the opt_in_TE_1550_device_wzwqoo_<label> text on each input grating coupler.

3. Design Equations and Values

Every device uses the 500 × 220 nm strip waveguide. Lumerical MODE gives $n_{\text{eff}} = 2.442$ and $n_g = 4.181$ at 1550 nm, within 0.24 % of the PDK model (4.191) [8]. The formulas below are from Chrostowski and Hochberg [1] unless another source is given.

Group index and drift. Any index change δn (from width, thickness or temperature) moves every filter by

$$n_g = n_{\text{eff}} - \lambda \frac{dn_{\text{eff}}}{d\lambda}, \quad \delta\lambda = \frac{\lambda \delta n}{n_g} \quad (1)$$

MZI and lattice delay. An unbalanced MZI with arm difference ΔL repeats every

$$\text{FSR} = \frac{\lambda^2}{n_g \Delta L} \quad (2)$$

Each lattice stage needs an FSR of twice the channel spacing it separates. Its couplers set the passband shape [2], [3]; a coupler of cross-power K needs a straight length L_c :

$$K = \sin^2 \left(\frac{\pi(L_c + L_0)}{2L_x} \right) \quad (3)$$

with the cross-over length L_x and the S-bend term L_0 taken from 3D FDTD of the PDK cell.

Ring. A ring of radius R has an FSR and a linewidth set by the power coupling κ^2 of each coupler [4]:

$$\text{FSR} = \frac{\lambda^2}{n_g 2\pi R}, \quad \text{FWHM} \approx \frac{\text{FSR} \kappa^2}{\pi} \quad (4)$$

CDC. Two guides of indices n_1, n_2 with a grating of period Λ drop one band backwards at [5]

$$\lambda_D = (n_1 + n_2) \Lambda \quad (5)$$

Bragg mirror and FP cavity. A grating of period Λ reflects at $\lambda_B = 2\bar{n}\Lambda$ with a peak reflection that grows with its length L :

$$\lambda_B = 2\bar{n}\Lambda, \quad R = \tanh^2(\kappa L) \quad (6)$$

Two mirrors with a half-period spacer give one narrow transmission line in the stop band [6].

AWG. The arm length step ΔL sets the grating order m and the FSR [7]:

$$m = \frac{n_{\text{eff}} \Delta L}{\lambda_c}, \quad \text{FSR} = \frac{\lambda_c^2}{n_g \Delta L} \quad (7)$$

Table 3. Design values used in the layout

Device	Parameter	Value
Lattice D1	Stage FSR $\rightarrow$ delay ΔL , by (2)	20 nm $\rightarrow$ 28.741 μm (stage 1); 40 nm $\rightarrow$ 14.370 / 14.530 μm (stages 2A / 2B)
	Coupler ratios $K \rightarrow$ straight lengths, by (3)	0.500, 0.0730, 0.0658, 0.500, 0.2154 $\rightarrow$ 17.106, 4.628, 4.284, 17.106, 9.724 μm
Rings D2	Radius, by (4)	7.5157 / 7.5289 / 7.5422 / 7.5555 μm for F1 / F2 / F3 / F4 (FSR $\approx$ 12.06 nm)
	Coupling for a 0.2 nm line, by (4)	$\kappa^2 = 0.049 \rightarrow$ gap 110 nm
CDC-A, CDC-B	Period, by (5); gap; length	$\Lambda = 316 / 318$ nm; gap 116 nm; 300 periods, uniform
FP cavity	Mirror, spacer	PDK Bragg grating $\Lambda = 317$ nm, 150 periods per mirror, 158 nm spacer
AWG-C	Order, arm step, arms, by (7)	$m = 11$, $\Delta L = 6.982$ μm , 27 arms
EDG-C	Order, radius, facets	$m = 14$, 168.9 μm grating circle, 28 facets with 3-pair Bragg mirrors
CDC cascade	Periods, length, apodisation	$\Lambda = 325 / 321 / 318$ nm (C4 / C3 / C2), 250 periods, Gaussian index $a = 1.3$
Chip 3	Width offsets	-15 and $+8$ nm (rings, CDC-A); -15 and $+15$ nm (MZIs)
	Matched ring	width $W^* = 479$ nm, $R = 7.639$ μm
	MZI delay	$\Delta L = 10$ μm , in 500 nm wire or in a 3 μm guide

4. Simulated Performance

Each component was solved in Lumerical (MODE FDE, EME, varFDTD or 3D FDTD), exported as S-parameters and assembled in INTERCONNECT with the PDK grating couplers. Table 4 is the expected measurement, with grating-coupler loss removed.

Table 4. Expected results per device

Label	Expected result
LatticeD1	Peak loss 0.36–0.49 dB; 1 dB band 8.4–8.9 nm; worst leak of another channel within ± 3 nm: $-25.3 / -21.8 / -19.1 / -15.1$ dB for C1 / C2 / C3 / C4
RingCDCD2	F1–F4 lines at 1542.93 / 1544.53 / 1546.11 / 1547.72 nm, 0.16–0.20 nm wide, peak -0.31 to -0.54 dB. Isolation about 30 dB on the CDC outputs, -24.1 dB on the direct ring drop
CDCA	1 dB band 1536.4–1545.0 nm, peak -0.05 dB at 1542.8 nm
GCLoopback	Two grating couplers: -4.71 dB peak at 1554.5 nm
AWGC	Peak -2.37 to -2.60 dB; 1 dB band 2.8–3.0 nm; worst leak within ± 3 nm -24.5 to -27.8 dB
EDGC	Peak -1.24 to -2.33 dB; isolation within ± 3 nm 19.9–32.3 dB
CDCCascade	Peak -0.09 to -0.16 dB; 1 dB band 7.0–8.1 nm; worst leak within ± 3 nm -16.5 to -20.2 dB
FPCavity	One line at 1544.28 nm, 0.30 nm wide, -0.33 dB (-5.51 dB through the two grating couplers)
Chip 3	See Table 6

Main comparisons.

- **Coarse filters.** The lattice has the widest flat top (8.4–8.9 nm at 1 dB) and low loss. The AWG and the EDG isolate better but have rounded passbands only 2.8–4.0 nm wide. The CDC cascade loses least but leaks most (−16.5 dB).
- **Fine filters.** A plain ring drop leaks −24.1 dB into the neighbouring channel, as (4) predicts for a 0.2 nm line. A CDC after the ring removes the ring's other orders, not this neighbour leak.
- **FP cavity.** Its 0.30 nm line rejects F1 by 20.5 dB, so it can separate two neighbouring fine channels that a CDC band (9 nm wide) cannot.

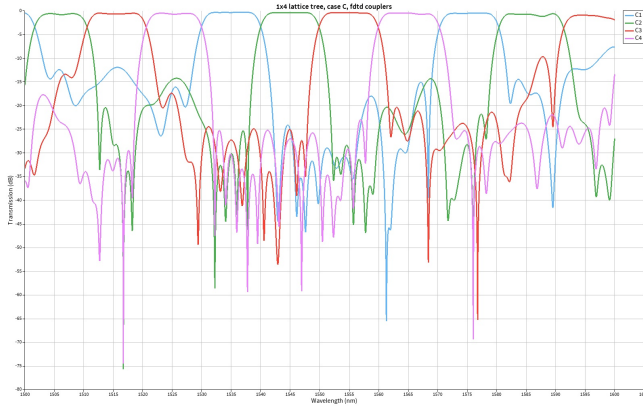


Fig. 2. Lattice tree D1, outputs C1–C4, couplers from 3D FDTD (INTERCONNECT).

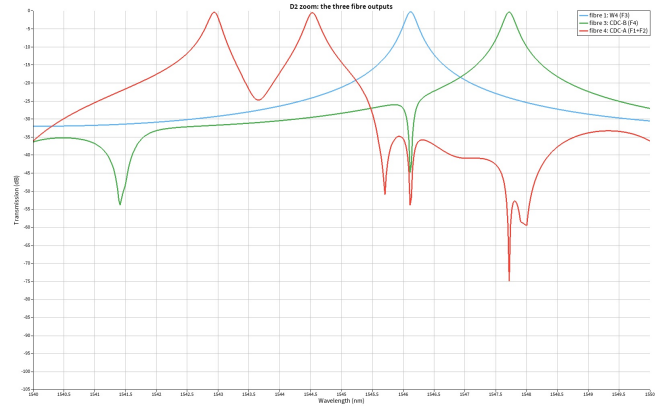


Fig. 3. Ring bank D2, 1540–1550 nm: W4 direct drop (fibre 1), CDC-B (fibre 3), CDC-A (fibre 4).

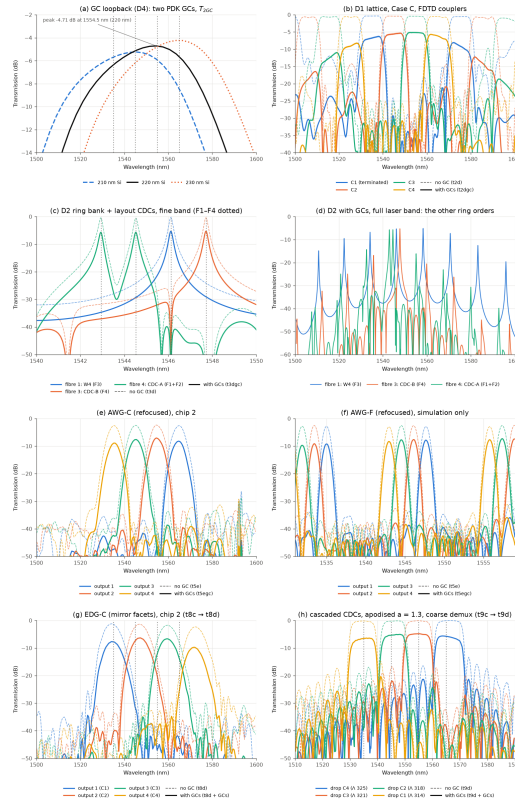


Fig. 4. What the fibre measurement will see: each device without (dashed) and with (solid) its two PDK grating couplers. (a) GC loopback at 210 / 220 / 230 nm Si. (b) Lattice D1. (c) Ring bank D2, fine band. (d) D2 over the full laser band. (e) AWG-C and (f) AWG-F (simulation only). (g) EDG-C. (h) CDC cascade.

5. Layout and Verification

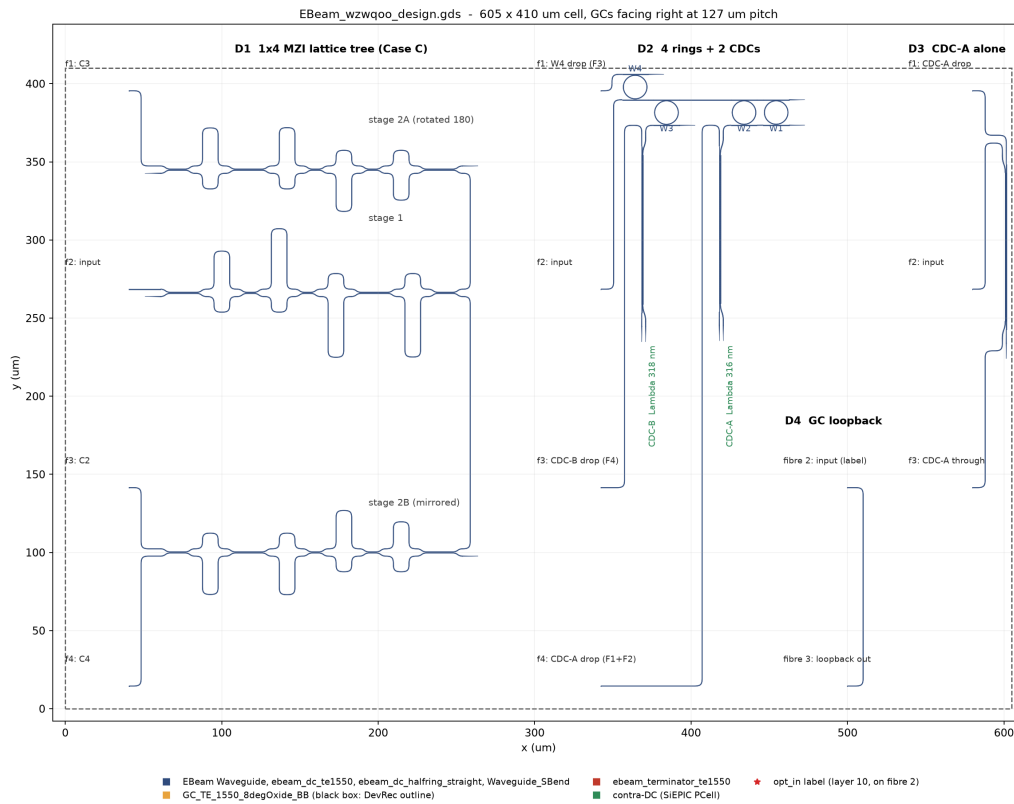


Fig. 5. Chip 1 (EBeam_wzwqoo_design). D1 lattice tree; D2 four rings and two CDCs; D3 CDC-A alone; D4 GC loopback.

Table 5. Verification of the submitted files (openEBL submission checks, SiEPIC verification and the EBeam DRC deck, run in KLayout on the files as submitted)

File = top cell	Si extent (μm)	Black-box GCs	DRC errors	Verification errors
EBeam_wzwqoo_design	561.2×392.0	13	0	0
EBeam_wzwqoo_benchmark	559.7×381.5	14	0	0
EBeam_wzwqoo_FabChallenges	534.1×386.3	20	0	0

Manufacturability choices. Every component is a PDK cell or PCell, with only its parameters changed:

- **CDC gap 116 nm instead of the default 100 nm.** At 100 nm the facing teeth are only 62.5 nm apart, below the 70 nm minimum Si spacing; at 116 nm they are 78.5 nm apart.
- **CDC-A and CDC-B unapodised.** The default apodisation tapers the end teeth to sub-nm depths, too shallow to print.
- **Cascade CDCs apodised at $a = 1.3$,** the strongest taper that keeps every tooth at least 5 nm deep (an assumed print limit, not a PDK rule).
- **Ring gap 110 nm and coupler gap 200 nm,** both above the 70 nm minimum.
- **FP cavity drawn as one straight piece,** so the spacer keeps its 158 nm length (a bend would stretch it and open extra lines).

Close waveguides. Two pairs of routing waveguides on chip 2 run 1.7–1.8 μm apart for about 21–24 μm . At that gap two 500 nm strips exchange a negligible fraction of their power, so no coupling is expected.

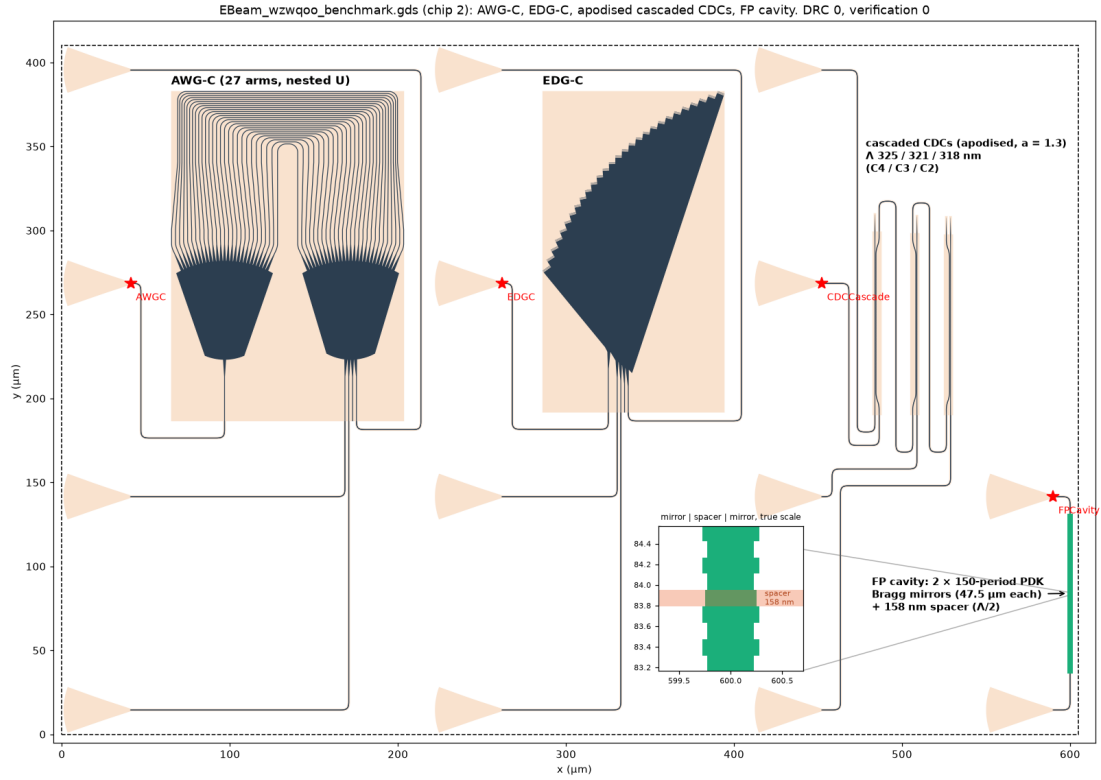


Fig. 6. Chip 2 (EBeam_wzwqoo_benchmark). AWG-C, EDG-C and three apodised cascaded CDCs, each with its own grating-coupler column, and the FP cavity at the right edge. Inset: the 158 nm spacer between the two Bragg mirrors at true scale.

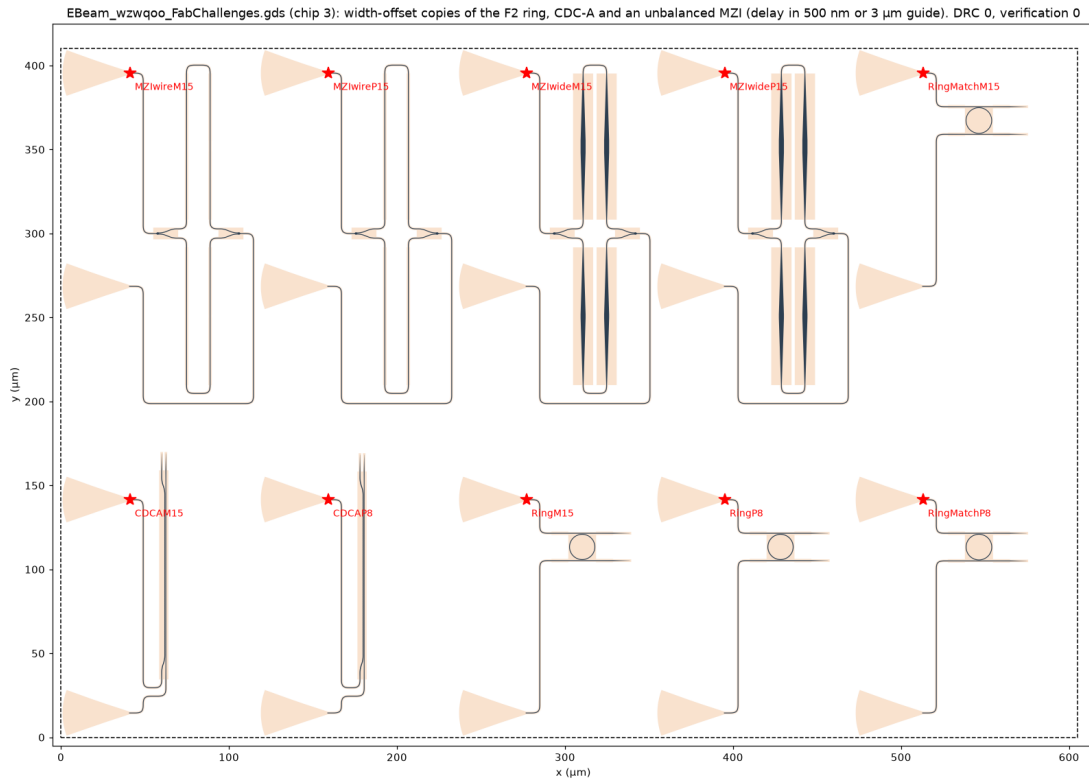


Fig. 7. Chip 3 (EBeam_wzwqoo_FabChallenges). Top: the two wire MZIs, the two 3 μm MZIs and the matched ring at -15 nm. Bottom: CDC-A at -15 and $+8$ nm, the 500 nm ring at -15 and $+8$ nm, and the matched ring at $+8$ nm.

6. Design for Test

- **One input, up to three outputs.** Every device is a column of PDK TE grating couplers (GC_TE_1550_8degOxide_BB) at 127 μm pitch. The opt_in label is on the input coupler; the other couplers of the column are the outputs.
- **Labels.** All 18 labels follow opt_in_TE_1550_device_wzwqoo_<label> and are unique across the three files (Table 2).
- **Unused ports** end in PDK terminators (ebeam_terminator_te1550).
- **Missing outputs.** With only three output fibres, D1 routes out C2, C3 and C4 (not C1), and the CDC cascade on chip 2 carries three CDCs (C4, C3, C2).
- **Calibration.** GCLoopback measures the grating-coupler pair; its spectrum is divided out of every other device. CDCA measures CDC-A alone, so the CDC band can be told apart from the ring lines in RingCDCD2. For chip 3, the F2 ring in RingCDCD2 and CDCA are the 0 nm references for the ring and CDC copies; the MZIs have no 0 nm copy and are read from the slope between their two copies.

Analysis plan.

1. Divide every spectrum by the GCLoopback spectrum.
2. LatticeD1, AWGC, EDGC, CDCCascade: read each output's peak, 1 dB band and worst leak within ± 3 nm, and compare them with Table 4.
3. RingCDCD2: fit each ring line (centre, FWHM), read n_g from the ring FSR by (4), and the isolation between outputs.
4. FPCavity: line position and width.
5. Chip 3: Section 7.

7. Fabrication and Temperature Experiments

Experiment 1: does the F2 ring stay inside CDC-A's band? On chip 1 the F2 line sits only 0.48 nm inside CDC-A's upper 1 dB edge. By (1), a width error d moves both, but at different rates: 0.591 nm per nm of width for the ring and 0.667 nm/nm for the CDC. The margin is used up at $d \approx -6$ nm. A ring 479 nm wide (W^*) moves at 0.662 nm/nm, nearly as fast as the CDC, and keeps F2 at every offset.

Experiment 2: a delay line that ignores width. In an MZI everything the two arms share (bends, splitters, equal straights) cancels. Only the extra length ΔL matters, so only the width of the guide that carries ΔL moves the fringes. Putting ΔL in a 3 μm guide instead of 500 nm wire makes the fringes 167 times less sensitive in simulation. $\Delta L = 10 \mu\text{m}$ gives an FSR of about 57 nm, so each spectrum shows one or two nulls. The experiment tracks the position of one null, which moves by less than half an FSR over the ± 15 nm range, so the same null is identified on every copy.

Table 6. Chip-3 predictions (INTERCONNECT)

Device	$d = -15$ nm	$d = 0$	$d = +8$ or $+15$ nm
MZI null, delay in 500 nm wire	1527.06 nm	1536.03 nm	1544.20 nm (+15)
MZI null, delay in 3 μm guide	1568.09 nm	1568.14 nm	1568.19 nm (+15)
CDC-A 1 dB band	1526.23–1534.83 nm	1536.40–1545.00 nm	1541.58–1550.18 nm (+8)
F2 ring line, 500 nm	1535.44 nm (outside the band: F2 lost)	1544.52 nm	1549.02 nm (+8)
F2 ring line, 479 nm (W^*)	1534.37 nm (0.46 nm inside)	1544.53 nm	1549.59 nm (+8)

The $d = 0$ column is simulated; the chip-1 devices provide it on the chip for the 500 nm ring and CDC-A only.

How chip 3 is read. The two copies of each device share the same unknown fabrication bias, so the slope between them (wavelength shift divided by the 23 or 30 nm width difference) does not depend on that bias.

Experiment 3: temperature. openEBL can measure devices a second time, about 10 K warmer. RingCDCD2, CDCA and FPCavity are requested. By (1), every feature should move by about +0.70 nm, and the ring and CDC by the same amount to within 0.005 nm. The measured shift gives the thermo-optic coefficient of the fabricated guide:

$$\frac{dn_{\text{eff}}}{dT} = \frac{n_g}{\lambda} \frac{\Delta\lambda}{\Delta T} \quad (8)$$

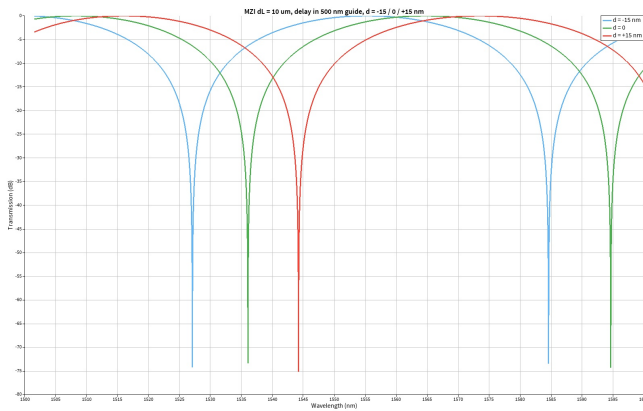


Fig. 8. MZI with its delay in 500 nm wire, $d = -15 / 0 / +15$ nm: the null moves about 9 nm per 15 nm of width.

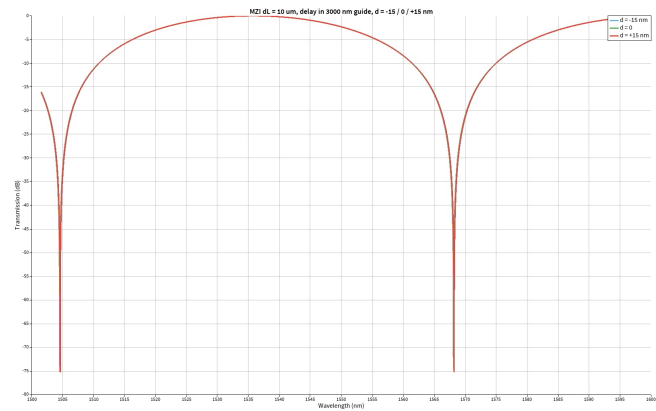


Fig. 9. The same MZI with its delay in a 3 μm guide: the three curves lie within 0.10 nm.

8. Test Results

Measurements pending: this section will hold the openEBL results.

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